EXPERIMENT-2

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AIM: Study the basic network command and Network configuration commands like ping,variations of ip config,tracert,nslookup,netstat,arp,rarp,hostname,pathping and basic networking commands.

- ping
- ipconfig
- tracert
- nslookup
- netstat
- arp
- hostname
- pathping
- route
- net use

OBJECTIVES:

- To study the basic Windows networking commands used for network configuration and troubleshooting.
- To understand the working of the Ping command and its variations for testing network connectivity using ICMP.
- To learn the use of IPConfig commands for viewing and managing TCP/IP configuration.

EXPERIMENT-2

- To understand the Tracert command for tracing the route taken by packets from the source to the destination.
- To study the NSLookup command for DNS name resolution and reverse lookup.
- To examine active network connections, listening ports, and protocol statistics using the Netstat command.
- To understand the ARP command for viewing and managing the ARP cache (IP-to-MAC address mapping).
- To learn the use of the Hostname, PathPing, Route, and Net Use commands for identifying the system, analyzing network paths, viewing routing information, etc.

SOFTWARE REQUIREMENTS:

(a) Software:

- Microsoft Windows 11 Operating System
- Windows Command Prompt (CMD)

(b) Hardware (Virtual):

- Desktop Computer or Laptop
- Network Interface Card (NIC)
- Internet/LAN Connection

THEORY: (i) Ping

Ping (Packet Internet Groper) is a network diagnostic utility used to verify whether a remote host or network device is reachable over an IP network. It works by sending **ICMP (Internet Control Message Protocol) Echo Request** packets to the destination host and waiting for **ICMP Echo Reply** packets. If the destination responds, the command displays the round-trip time (RTT), packet loss, and Time To Live (TTL) value. Ping is one of the most commonly used tools for testing network connectivity and troubleshooting communication problems.

Working of Ping

1. The source computer sends an ICMP Echo Request packet to the destination.
2. The destination receives the packet and sends back an ICMP Echo Reply.
3. The source measures the time taken for the packet to travel to the destination and back.

EXPERIMENT-2

4. If no reply is received within a specified time, the request is considered timed out.

Purpose of Ping

- To verify whether a host is reachable.
- To test network connectivity.
- To measure response time (latency).
- To detect packet loss.
- To troubleshoot network problems.

Common Variations

- ping <IP/hostname> – Tests connectivity.
- ping -t – Sends continuous Echo Requests until stopped.
- ping -n <count> – Sends a specified number of Echo Requests.
- ping -l <size> – Specifies the size of the data payload.
- ping -f – Sets the "Don't Fragment" flag.
- ping -a – Resolves an IP address into its hostname.

(ii) IPConfig

IPConfig (Internet Protocol Configuration) is a Windows command-line utility used to display and manage the TCP/IP configuration of network interfaces. It provides information such as the IP address, subnet mask, default gateway, DHCP status, and DNS server details. It is an essential command for verifying network configuration and troubleshooting IP-related issues.

Purpose

- Display network configuration.
- Verify IP address assignment.
- Check subnet mask and gateway.

- Renew or release DHCP addresses.
- Troubleshoot IP connectivity problems.

(iii) Tracert

Tracert (Trace Route) is a Windows networking command that traces the complete path taken by packets from the source computer to the destination. Unlike Ping, which only confirms whether a destination is reachable, Tracert identifies every intermediate router (hop) through which the packets travel. It uses ICMP Echo Request packets with increasing TTL (Time-To-Live) values.

Working

- Sends packets with TTL = 1.
- The first router decreases TTL to zero and returns an ICMP Time Exceeded message.
- TTL is gradually increased to discover each router along the path.
- The process continues until the destination is reached.

Purpose

- Display the routing path.
- Identify slow routers.
- Detect routing failures.
- Troubleshoot Internet connectivity.

(iv) NSLookup

NSLookup (Name Server Lookup) is a command-line utility used to query the Domain Name System (DNS). DNS translates human-readable domain names into IP addresses. NSLookup helps verify DNS functionality and diagnose DNS-related issues.

Purpose

- Resolve domain names to IP addresses.

EXPERIMENT-2

- Perform reverse DNS lookup.
- Verify DNS server operation.
- Retrieve different DNS record types.

(v) Netstat

Netstat (Network Statistics) is a command-line utility that displays network-related information such as active connections, listening ports, routing tables, protocol statistics, and Ethernet statistics. It is widely used by network administrators to monitor network activity and identify suspicious or unnecessary connections.

Purpose

- View active TCP/UDP connections.
- Display listening ports.
- Show routing information.
- Display protocol statistics.
- Identify process IDs using network ports.

(vi) ARP

ARP (Address Resolution Protocol) is used to map an IPv4 address to its corresponding MAC address on a local network. Communication over Ethernet requires MAC addresses, while applications use IP addresses. ARP bridges this gap by maintaining an **ARP Cache**.

Working

1. The sender knows the destination IP.
2. It broadcasts an ARP Request.
3. The destination replies with its MAC address.
4. The sender stores the mapping in the ARP cache.

Purpose

EXPERIMENT-2

- View ARP cache.
- Delete ARP entries.
- Create static ARP entries.
- Troubleshoot LAN communication.

(vii) Hostname

The **Hostname** command displays the name assigned to the current computer. Every computer connected to a network has a unique hostname for easy identification.

Purpose

- Identify the computer on the network.
- Assist in network administration.
- Verify the system name.

(viii) PathPing

PathPing combines the features of **Ping** and **Tracert**. It traces the complete path to the destination and then sends multiple ICMP Echo Requests to each router to calculate packet loss and network latency at every hop.

Purpose

- Detect packet loss.
- Identify slow routers.
- Analyze network latency.
- Troubleshoot network performance.

Working

1. Finds the route like Tracert.
2. Sends multiple ping packets to every hop.

EXPERIMENT-2

3. Calculates packet loss percentage.
4. Displays latency and delay.

(ix) Route

The **Route** command is used to display and manage the routing table in Windows. A routing table contains information about the paths used to forward packets to different networks.

Purpose

- Display routing table.
- Add static routes.
- Delete routes.
- Modify routing entries.

(x) Net Use

Net Use is a Windows networking command used to connect, disconnect, and manage shared network resources such as shared folders and printers. It allows users to map a network share to a drive letter for easier access.

Purpose

- Display shared network resources.
- Map network drives.
- Disconnect mapped drives.
- Manage network connections.

STEPS/IMPLEMENTATION:

Part I – Ping Command

EXPERIMENT-2

1. Open the Command Prompt (CMD) by pressing Windows + R, typing cmd, and pressing Enter.
2. Execute the Ping command to check network connectivity.
3. Observe the reply time, TTL value, and packet loss.
4. Execute different Ping variations such as -t, -n, -l, -f, and -a to understand their functions.

Part II – IPConfig Command

6. Execute the ipconfig command to view the current IP configuration.
7. Execute ipconfig /all to display detailed TCP/IP configuration.
8. Execute ipconfig /release to release the current DHCP-assigned IP address.
9. Execute ipconfig /renew to obtain a new IP address from the DHCP server.
10. Observe the IP address, subnet mask, default gateway, DNS server, and MAC address.

Part III – Tracert Command

11. Execute the tracert command to trace the route to a destination.
12. Observe the number of hops, router IP addresses, and response time for each hop.
13. Execute tracert -d to display only IP addresses.
14. Execute tracert -h to specify the maximum number of hops.

Part IV – NSLookup Command

15. Execute the nslookup command to resolve a domain name into an IP address.
16. Perform reverse DNS lookup using an IP address.
17. Query different DNS record types such as **A** and **AAAA** records.
18. Verify DNS server functionality.

EXPERIMENT-2

Part V – Netstat Command

19. Execute the netstat command to display active network connections.
20. Use netstat -a to display all active and listening ports.
21. Execute netstat -n, netstat -o, netstat -e, netstat -r, netstat -s, and netstat -ano.
22. Observe protocol statistics, routing information, Ethernet statistics, and Process IDs (PID).

Part VI – ARP Command

23. Execute the arp -a command to display the ARP cache.
24. Observe the IP-to-MAC address mappings.
25. Delete ARP entries using arp -d.
26. Create a static ARP entry using arp -s.

Part VII – Hostname Command

27. Execute the hostname command.
28. Observe the current computer name used on the network.

Part VIII – PathPing Command

29. Execute the pathping command for a remote host.
30. Observe the complete route, packet loss percentage, and latency at each router.
31. Identify any router causing delays or packet loss.

Part IX – Route Command

EXPERIMENT-2

32. Execute the route print command to display the routing table.
33. Add a static route using the route add command.
34. Delete the static route using the route delete command.
35. Observe the changes in the routing table.

Part X – Net Use Command

36. Execute the net use command to display currently connected network resources.
37. Map a shared folder to a drive letter using the net use command.
38. Disconnect the mapped drive using net use /delete.
39. Verify that the network resource has been removed successfully.

DEMONSTRATION/OUTPUT:

Part-1: Ping command

```
C:\Users\husan>ping google.com

Pinging google.com [142.250.118.113] with 32 bytes of data:
Reply from 142.250.118.113: bytes=32 time=19ms TTL=112
Reply from 142.250.118.113: bytes=32 time=18ms TTL=112
Reply from 142.250.118.113: bytes=32 time=18ms TTL=112
Reply from 142.250.118.113: bytes=32 time=19ms TTL=112

Ping statistics for 142.250.118.113:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 19ms, Average = 18ms
```

```
C:\Users\husan>ping -t google.com

Pinging google.com [192.178.134.102] with 32 bytes of data:
Reply from 192.178.134.102: bytes=32 time=37ms TTL=109
Reply from 192.178.134.102: bytes=32 time=37ms TTL=109
Reply from 192.178.134.102: bytes=32 time=36ms TTL=109
Reply from 192.178.134.102: bytes=32 time=37ms TTL=109
Reply from 192.178.134.102: bytes=32 time=37ms TTL=109
Reply from 192.178.134.102: bytes=32 time=36ms TTL=109
Reply from 192.178.134.102: bytes=32 time=36ms TTL=109
Reply from 192.178.134.102: bytes=32 time=36ms TTL=109

Ping statistics for 192.178.134.102:
    Packets: Sent = 8, Received = 8, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 36ms, Maximum = 37ms, Average = 36ms
```

EXPERIMENT-2

```
C:\Users\husan>ping -n 5 192.178.134.102

Pinging 192.178.134.102 with 32 bytes of data:
Reply from 192.178.134.102: bytes=32 time=45ms TTL=109
Reply from 192.178.134.102: bytes=32 time=47ms TTL=109
Reply from 192.178.134.102: bytes=32 time=44ms TTL=109
Reply from 192.178.134.102: bytes=32 time=38ms TTL=109
Reply from 192.178.134.102: bytes=32 time=40ms TTL=109

Ping statistics for 192.178.134.102:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 38ms, Maximum = 47ms, Average = 42ms
```

```
C:\Users\husan>ping -l 100 192.178.134.102

Pinging 192.178.134.102 with 100 bytes of data:
Reply from 192.178.134.102: bytes=100 time=45ms TTL=109
Reply from 192.178.134.102: bytes=100 time=39ms TTL=109
Reply from 192.178.134.102: bytes=100 time=37ms TTL=109
Reply from 192.178.134.102: bytes=100 time=39ms TTL=109

Ping statistics for 192.178.134.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 37ms, Maximum = 45ms, Average = 40ms
```

```
C:\Users\husan>ping -f -l 1472 google.com

Pinging google.com [142.250.193.14] with 1472 bytes of data:
Reply from 142.250.193.14: bytes=1472 time=10ms TTL=113
Reply from 142.250.193.14: bytes=1472 time=12ms TTL=113
Reply from 142.250.193.14: bytes=1472 time=11ms TTL=113
Reply from 142.250.193.14: bytes=1472 time=11ms TTL=113

Ping statistics for 142.250.193.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 12ms, Average = 11ms
```

```
C:\Users\husan>ping -a 192.178.134.102

Pinging lcdely-in-f102.1e100.net [192.178.134.102] with 32 bytes of data:
Reply from 192.178.134.102: bytes=32 time=17ms TTL=110
Reply from 192.178.134.102: bytes=32 time=8ms TTL=110
Reply from 192.178.134.102: bytes=32 time=9ms TTL=110
Reply from 192.178.134.102: bytes=32 time=10ms TTL=110

Ping statistics for 192.178.134.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 17ms, Average = 11ms
```

Part-2: IPConfig Command

EXPERIMENT-2

```
C:\Users\husan>ipconfig

Windows IP Configuration


Ethernet adapter vEthernet (WSLCore):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::fab4:102d:432e:3f8f%72
   IPv4 Address. . . . . . . . . . . : 172.18.224.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Unknown adapter McAfee VPN:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Unknown adapter OpenVPN TAP-Windows6:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

```
C:\Users\husan>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Husanpreet
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : Yes
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : cuchdit.in

Ethernet adapter vEthernet (WSLCore):

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Hyper-V Virtual Ethernet Adapter
   Physical Address. . . . . . . . . : 00-15-5D-79-B4-C2
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::fab4:102d:432e:3f8f%72(Preferred)
   IPv4 Address. . . . . . . . . . . : 172.18.224.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 1207965021
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2F-90-62-9A-B8-1E-A4-D6-A0-31
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

```
C:\Users\husan>ipconfig /release

Windows IP Configuration

No operation can be performed on McAfee VPN while it has its media disconnected.
No operation can be performed on OpenVPN TAP-Windows6 while it has its media disconnected.
No operation can be performed on Local Area Connection while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 12 while it has its media disconnected.

Ethernet adapter vEthernet (WSLCore):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::fab4:102d:432e:3f8f%72
   IPv4 Address. . . . . . . . . . . : 172.18.224.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :
```

EXPERIMENT-2

```
C:\Users\husan>ipconfig /renew

Windows IP Configuration

No operation can be performed on McAfee VPN while it has its media disconnected.
No operation can be performed on OpenVPN TAP-Windows6 while it has its media disconnected.
No operation can be performed on OpenVPN Data Channel Offload while it has its media disconnected.
No operation can be performed on Local Area Connection while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 12 while it has its media disconnected.

Ethernet adapter vEthernet (WSLCore):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::fab4:102d:432e:3f8f%72
   IPv4 Address. . . . . . . . . . . : 172.18.224.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :
```

Part-3: Tracert Command

```
C:\Users\husan>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1    23 ms     3 ms    23 ms  dlinkrouter.cuchdit.in [192.168.0.1]
  2     7 ms     6 ms     7 ms  172.20.56.1
  3    14 ms     5 ms     3 ms  192.168.40.2
  4    12 ms     4 ms     2 ms  172.16.2.1
  5    11 ms     3 ms     4 ms  static-145.148.249.49-tataidc.co.in [49.249.148.145]
  6    19 ms    11 ms    10 ms  10.98.30.233
  7    29 ms     9 ms    11 ms  10.129.17.17
  8     *        *        *     Request timed out.
  9     *        *       10 ms  10.129.33.117
 10    17 ms     9 ms     8 ms  72.14.196.97
 11    27 ms     9 ms     9 ms  142.251.77.185
 12    32 ms    13 ms     9 ms  209.85.252.65
 13    23 ms    12 ms    12 ms  dns.google [8.8.8.8]

Trace complete.
```

```
C:\Users\husan>tracert -d 8.8.8.8

Tracing route to 8.8.8.8 over a maximum of 30 hops

  1    11 ms     3 ms     2 ms  192.168.0.1
  2    14 ms     5 ms     5 ms  172.20.56.1
  3     7 ms     6 ms     3 ms  192.168.40.2
  4     3 ms     2 ms    10 ms  172.16.2.1
  5    25 ms    19 ms    39 ms  112.196.95.1
  6     5 ms     6 ms     6 ms  202.164.32.25
  7    10 ms    17 ms     5 ms  202.164.51.36
  8    15 ms    14 ms    15 ms  72.14.213.36
  9    17 ms    18 ms    19 ms  142.251.77.23
 10    18 ms    18 ms    42 ms  142.251.49.115
 11    14 ms    14 ms    14 ms  8.8.8.8

Trace complete.
```

EXPERIMENT-2

```
C:\Users\husan>tracert -h 5 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 5 hops:

  1    25 ms     2 ms     3 ms  dlinkrouter.cuchdit.in [192.168.0.1]
  2     6 ms     5 ms     6 ms  172.20.56.1
  3    28 ms     2 ms    13 ms  192.168.40.2
  4    14 ms    14 ms    11 ms  172.16.2.1
  5    31 ms    21 ms    20 ms  112.196.95.1

Trace complete.
```

```
C:\Users\husan>tracert -?

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
               [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d                 Do not resolve addresses to hostnames.
    -h maximum_hops    Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                 Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                 Force using IPv4.
    -6                 Force using IPv6.
```

Part-4: NSLookup Command

```
C:\Users\husan>nslookup google.com
Server:  dlinkrouter.cuchdit.in
Address:  192.168.0.1

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4013:802::64
          2404:6800:4013:802::66
          2404:6800:4013:802::8a
          2404:6800:4013:802::8b
          142.250.183.238
```

```
C:\Users\husan>nslookup -type=A google.com
Server:  dlinkrouter.cuchdit.in
Address:  192.168.0.1

Non-authoritative answer:
Name:    google.com
Addresses:  192.178.134.138
          192.178.134.102
          192.178.134.100
          192.178.134.101
          192.178.134.113
          192.178.134.139
```

EXPERIMENT-2

Part-5: Netstat Command

```
C:\Users\husan>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:4316         kubernetes:4317        ESTABLISHED
  TCP    127.0.0.1:4317         kubernetes:4316        ESTABLISHED
  TCP    127.0.0.1:5354         kubernetes:49673       ESTABLISHED
  TCP    127.0.0.1:5354         kubernetes:49678       ESTABLISHED
  TCP    127.0.0.1:27015        kubernetes:49774       ESTABLISHED
  TCP    127.0.0.1:45595        kubernetes:49679       TIME_WAIT
  TCP    127.0.0.1:45596        kubernetes:49679       TIME_WAIT
  TCP    127.0.0.1:45597        kubernetes:49752       TIME_WAIT
  TCP    127.0.0.1:45598        kubernetes:49679       TIME_WAIT
  TCP    127.0.0.1:45599        kubernetes:49679       TIME_WAIT
  TCP    127.0.0.1:45600        kubernetes:49679       TIME_WAIT
  TCP    127.0.0.1:45605        kubernetes:49679       TIME_WAIT
  TCP    127.0.0.1:49673        kubernetes:5354        ESTABLISHED
  TCP    127.0.0.1:49678        kubernetes:5354        ESTABLISHED
  TCP    127.0.0.1:49679        kubernetes:45595       TIME_WAIT
  TCP    127.0.0.1:49679        kubernetes:45598       TIME_WAIT
```

```
C:\Users\husan>netstat -o

Active Connections

  Proto  Local Address          Foreign Address        State           PID
  TCP    127.0.0.1:4316         kubernetes:4317        ESTABLISHED     16960
  TCP    127.0.0.1:4317         kubernetes:4316        ESTABLISHED     25288
  TCP    127.0.0.1:5354         kubernetes:49673       ESTABLISHED     4824
  TCP    127.0.0.1:5354         kubernetes:49678       ESTABLISHED     4824
  TCP    127.0.0.1:17400        kubernetes:49679       TIME_WAIT       0
  TCP    127.0.0.1:17402        kubernetes:49679       TIME_WAIT       0
  TCP    127.0.0.1:17403        kubernetes:49679       TIME_WAIT       0
  TCP    127.0.0.1:27015        kubernetes:49774       ESTABLISHED     4796
  TCP    127.0.0.1:49673        kubernetes:5354        ESTABLISHED     4796
  TCP    127.0.0.1:49678        kubernetes:5354        ESTABLISHED     4796
  TCP    127.0.0.1:49679        kubernetes:49680       TIME_WAIT       0
  TCP    127.0.0.1:49679        kubernetes:49693       TIME_WAIT       0
  TCP    127.0.0.1:49679        kubernetes:49694       TIME_WAIT       0
  TCP    127.0.0.1:49679        kubernetes:49696       TIME_WAIT       0
```

```
C:\Users\husan>netstat -r
===========================================================================
Interface List
 11...00 ff 2c 72 d6 7d ......TAP-Windows Adapter V9 #2
 26...00 ff de 8a ef fb ......TAP-Windows Adapter V9
  6...........................OpenVPN Data Channel Offload
  5...00 ff 09 03 b5 9d ......TAP-ProtonVPN Windows Adapter V9
 20...ba 1e a4 d6 a0 31 ......Microsoft Wi-Fi Direct Virtual Adapter
 15...be 1e a4 d6 a0 31 ......Microsoft Wi-Fi Direct Virtual Adapter #4
 16...00 50 56 c0 00 01 ......VMware Virtual Ethernet Adapter for VMnet1
 25...00 50 56 c0 00 08 ......VMware Virtual Ethernet Adapter for VMnet8
 10...b8 1e a4 d6 a0 31 ......Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
  1...........................Software Loopback Interface 1
 70...00 15 5d cf c1 02 ......Hyper-V Virtual Ethernet Adapter
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.0.1    192.168.0.128     40
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
     172.17.160.0    255.255.240.0         On-link      172.17.160.1   5256
     172.17.160.1  255.255.255.255         On-link      172.17.160.1   5256
   172.17.175.255  255.255.255.255         On-link      172.17.160.1   5256
```

EXPERIMENT-2

```
C:\Users\husan>netstat -e
Interface Statistics

                           Received            Sent

Bytes                     405653209       136485926
Unicast packets              453865          223584
Non-unicast packets             504            4227
Discards                          0               0
Errors                            0               0
Unknown protocols                 0
```

Part-6: ARP Command

```
C:\Users\husan>arp -a

Interface: 192.168.0.128 --- 0xa
  Internet Address      Physical Address      Type
  192.168.0.1           c8-78-7d-6d-c0-1d     dynamic
  192.168.0.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  228.8.8.8             01-00-5e-08-08-08     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.193.1 --- 0x10
  Internet Address      Physical Address      Type
  192.168.193.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  228.8.8.8             01-00-5e-08-08-08     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Part-7: Hostname

```
C:\Users\husan>hostname
Husanpreet
```

Part-8: Pathping Command

EXPERIMENT-2

```
C:\Users\husan>pathping google.com

Tracing route to google.com [192.178.173.138]
over a maximum of 30 hops:
  0  Husanpreet.cuchdit.in [192.168.0.128]
  1  dlinkrouter.cuchdit.in [192.168.0.1]
  2  172.20.56.1
  3  192.168.40.2
  4  172.16.2.1
  5  static-145.148.249.49-tataidc.co.in [49.249.148.145]
  6  10.98.30.233
  7  10.129.17.17
  8     *        *        *
Computing statistics for 175 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           Husanpreet.cuchdit.in [192.168.0.128]
                                0/ 100 =  0%   |
  1    6ms     0/ 100 =  0%     0/ 100 =  0%  dlinkrouter.cuchdit.in [192.168.0.1]
                                0/ 100 =  0%   |
  2    9ms     0/ 100 =  0%     0/ 100 =  0%  172.20.56.1
                                0/ 100 =  0%   |
  3    8ms     0/ 100 =  0%     0/ 100 =  0%  192.168.40.2
                                0/ 100 =  0%   |
  4    5ms     0/ 100 =  0%     0/ 100 =  0%  172.16.2.1
                                0/ 100 =  0%   |
  5    8ms     0/ 100 =  0%     0/ 100 =  0%  static-145.148.249.49-tataidc.co.in [49.249.148.145]
                              100/ 100 =100%   |
  6  ---     100/ 100 =100%     0/ 100 =  0%  10.98.30.233
                                0/ 100 =  0%   |
  7  ---     100/ 100 =100%     0/ 100 =  0%  10.129.17.17

Trace complete.
```

Part-9: Route Command

```
C:\Users\husan>route print
===========================================================================
Interface List
 11...00 ff 2c 72 d6 7d ......TAP-Windows Adapter V9 #2
 26...00 ff de 8a ef fb ......TAP-Windows Adapter V9
  6...........................OpenVPN Data Channel Offload
  5...00 ff 09 03 b5 9d ......TAP-ProtonVPN Windows Adapter V9
 20...ba 1e a4 d6 a0 31 ......Microsoft Wi-Fi Direct Virtual Adapter
 15...be 1e a4 d6 a0 31 ......Microsoft Wi-Fi Direct Virtual Adapter #4
 16...00 50 56 c0 00 01 ......VMware Virtual Ethernet Adapter for VMnet1
 25...00 50 56 c0 00 08 ......VMware Virtual Ethernet Adapter for VMnet8
 10...b8 1e a4 d6 a0 31 ......Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
  1...........................Software Loopback Interface 1
 70...00 15 5d cf c1 02 ......Hyper-V Virtual Ethernet Adapter
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.0.1    192.168.0.128     40
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
     172.17.160.0    255.255.240.0         On-link      172.17.160.1   5256
     172.17.160.1  255.255.255.255         On-link      172.17.160.1   5256
   172.17.175.255  255.255.255.255         On-link      172.17.160.1   5256
      192.168.0.0    255.255.255.0         On-link     192.168.0.128    296
    192.168.0.128  255.255.255.255         On-link     192.168.0.128    296
```

Part-10: Net use Command

```
C:\Users\husan>net use
New connections will be remembered.

There are no entries in the list.
```

LEARNING OUTCOMES:

EXPERIMENT-2

After completing this experiment:

- I understood the purpose and working of essential Windows networking commands such as Ping, IPConfig, Tracert, NSLookup, Netstat, ARP, Hostname, PathPing, Route, and Net Use, and learned how they are used for network configuration, monitoring, and troubleshooting.
- I gained practical knowledge of network connectivity testing by using the Ping command and its various options to verify communication between devices, measure network latency, identify packet loss, and diagnose basic network connectivity issues.
- I learned to view and manage TCP/IP network configuration using IPConfig, trace packet routes using Tracert, resolve domain names using NSLookup, and analyze routing paths and packet loss using PathPing, enabling effective diagnosis of network communication problems.
- I developed the ability to monitor and analyze network activity using Netstat, understand IP-to-MAC address mapping through ARP, identify the system using the Hostname command, and manage routing information and shared network resources using the Route and Net Use commands.
- I enhanced my practical skills in network administration and troubleshooting by executing command-line networking utilities in Windows, interpreting their outputs, identifying common network issues, and applying appropriate commands to verify connectivity, resolve configuration problems, and manage network resources efficiently.